# AIRCRAFT SYSTEMS

## LANDING GEAR

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## DSC-32-10 Gears and Doors

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| Localization Title | Toc Index | ID | Reason |
|---|---|---|---|
| DSC-32-10-30<br>Proximity Detectors on Shock Absorbers | B | 1 | Documentation update: Deletion of text. |
| | | | Removal of non applicable footnote. |
| | | | Documentation update: Deletion of information. |
| | | | Update of the CAS value for the closure of the safety valve: 264 kt replaces 260 kt. |
| DSC-32-30-10<br>Auto Brake - System Arming | C | 1 | Effectivity update: The information now also applies to MSN 05004, 12883. |
| DSC-32-30-10<br>Auto Brake - System Arming | C | 2 | Effectivity update: The information no longer applies to MSN 05004. |
| DSC-32-30-10<br>Auto Brake - System Activation | C | 3 | Effectivity update: The information now also applies to MSN 05004, 12883. |
| DSC-32-30-10<br>Auto Brake - System Activation | C | 4 | Effectivity update: The information no longer applies to MSN 05004. |
| DSC-32-30-20<br>Auto BRK Panel | B | 1 | Effectivity update: The information now also applies to MSN 12883. |
| DSC-32-30-20<br>Audio Indicators | F | 2 | Effectivity update: The information now also applies to MSN 12883. |
| DSC-32-30-20<br>AUTO BRK OFF | G | 3 | Effectivity update: The information now also applies to MSN 12883. |

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## GENERAL

Ident.: DSC-32-10-10-00018598.0001001 / 21 MAR 16
**Applicable to: ALL**

The landing gear consists of :
- Two main landing gears that retract inboard
- One nose landing gear that retracts forward.

Doors enclose the landing gear bays. Gear and doors are electrically controlled and hydraulically operated.
The doors, which are fitted to the landing gear struts, are operated mechanically by the gear and close at the end of gear retraction.
All gear doors open while the gear is retracting or extending.
Two Landing Gear Control and Interface Units (LGCIUs) control the extension and retraction of the gear and the operation of the doors. They also supply information about the landing gear to ECAM for display, and send signals indicating whether the aircraft is in flight or on the ground to other aircraft systems.
A hand crank on the center pedestal allows the flight crew to extend the landing gear if the aircraft loses hydraulic systems or electrical power.

## MAIN LANDING GEAR (MLG)

Ident.: DSC-32-10-10-00018599.0001001 / 21 MAR 16
**Applicable to: ALL**

Each main gear has twin wheels and an oleopneumatic shock absorber.
Each main wheel has an antiskid brake.

## NOSE LANDING GEAR (NLG)

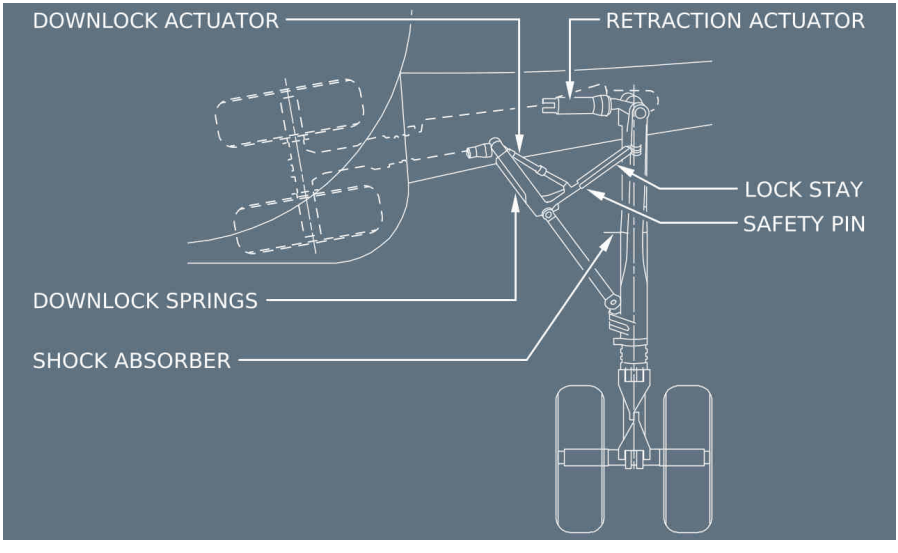
Ident.: DSC-32-10-10-00018600.0001001 / 21 MAR 16
**Applicable to: ALL**

The two-wheeled nose gear has an oleopneumatic shock strut and a nose wheel steering system.

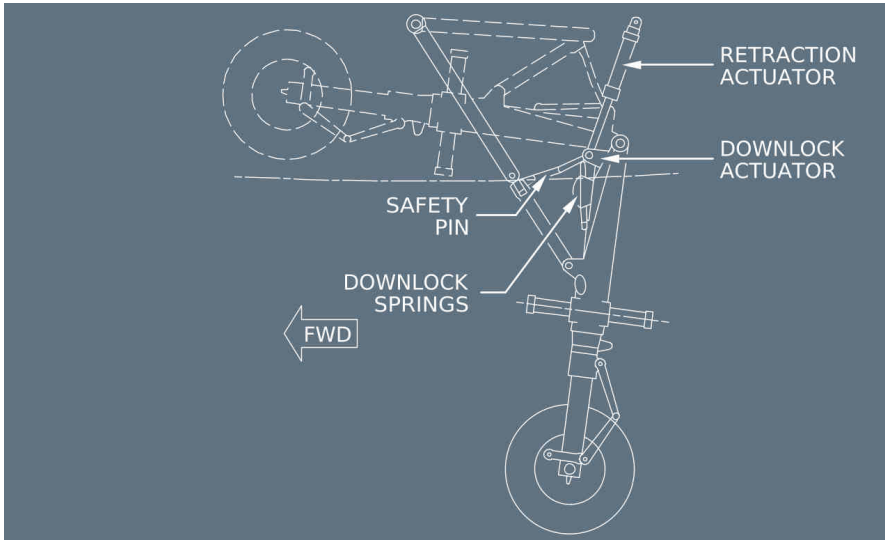
## LANDING GEAR EXTENSION AND RETRACTION EQUIPMENT

dent.: DSC-32-10-10-00018601.0001001 / 21 MAR 16
**Applicable to: ALL**

### MAIN LANDING GEAR



## NOSE LANDING GEAR



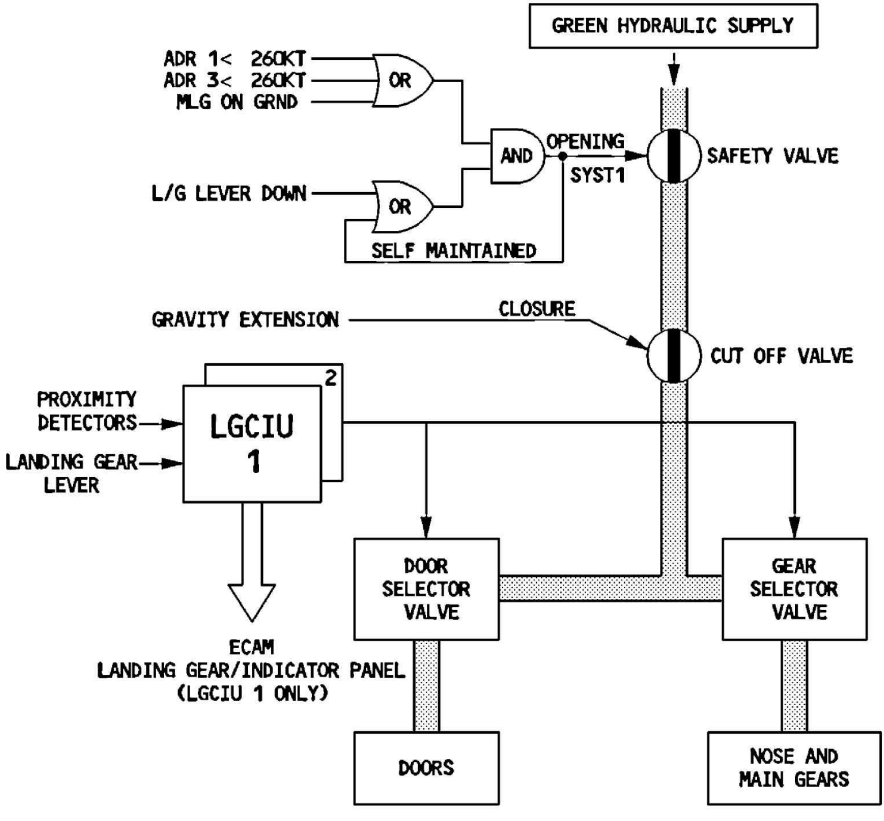

## LANDING GEARS AND DOORS OPERATION

**Applicable to: ALL**

Ident.: DSC-32-10-10-A-00018602.0001001 / 18 JUL 24

## NORMAL OPERATION

The flight crew normally operates the landing gear by means of the lever on the center instrument panel.
The LGCIUs control the sequencing of gear and doors electrically. One LGCIU controls one complete gear cycle, then switches over automatically to the other LGCIU at the completion of the retraction cycle. It also switches over in case of failure.
The green hydraulic system actuates all gear and doors. When the aircraft speed is above 264 kt, a safety valve automatically cuts off hydraulic supply to the landing gear system. Below 260 kt, the hydraulic supply remains cut off as long as the landing gear lever is up.




dent.: DSC-32-10-10-A-00018603.0001001 / 21 MAR 16

## LANDING GEAR GRAVITY EXTENSION

If the normal system fails to extend the landing gear hydraulically, the flight crew can use a crank to extend it mechanically.

When a crew member turns the crank, it :
- Isolates the landing gear hydraulics from the green hydraulic system
- Unlocks the landing gear doors and the main and nose main gear
- Allows gravity to drop the gear into the extended position.

Locking springs help the crew to crank the main gear into the locked condition, and aerodynamic forces assist in the locking of the nose gear.

The gear doors remain open.

The flight crew can reset the emergency extension system in flight after using it for training (if green hydraulic pressure is available).

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# LANDING GEAR CONTROL INTERFACE UNIT (LGCIU)

**Applicable to: ALL**

Ident.: DSC-32-10-20-A-00018604.0001001 / 21 MAR 16

## GENERAL

The LGCIUs receive position information from the landing gear, cargo door, and landing flap systems.

Ident.: DSC-32-10-20-A-00018605.0001001 / 21 MAR 16

## LANDING GEAR INFORMATION

The LGCIUs receive landing gear position information from proximity detectors when:
- The landing gears are locked down or up, or
- The shock absorbers are compressed or extended, or
- The landing gear doors are open, or closed, or
- The bogie are aligned or not.

The LGCIUs send the landing gear position data to other aircraft systems.
In case of a LGCIU failure, the landing gear is controlled by the remaining healthy LGCIU.

Ident.: DSC-32-10-20-A-00018606.0001001 / 17 MAY 21

## CARGO DOORS INFORMATION

Sensors send to the LGCIUs the position of the following components :
- Manual selector valves
- Locking shaft
- Locking handle
- Safety shaft
- Door sills .

The LGCIUs detect electrical failures only in certain proximity switches in the cargo door system :
- Locking shaft
- Locking handle
- Safety shaft.

When an LGCIU makes such a detection, it indicates the NON LOCKED condition for that component.

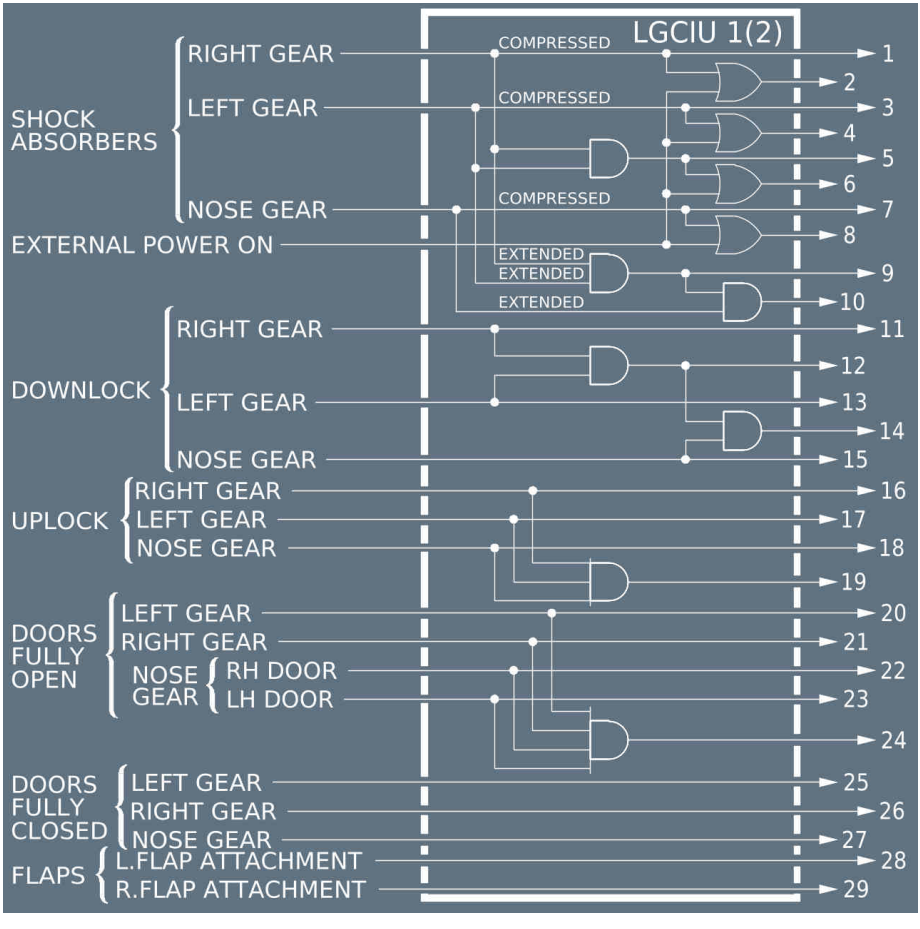
Ident.: DSC-32-10-20-A-00018607.0001001 / 21 MAR 16

## LANDING FLAPS INFORMATION

The LGCIUs process the signals from four flap disconnect proximity switches, then send them to the Slat/Flap Control Computers (SFCCs).
The LGCIUs do not monitor failures in the SFCC system.

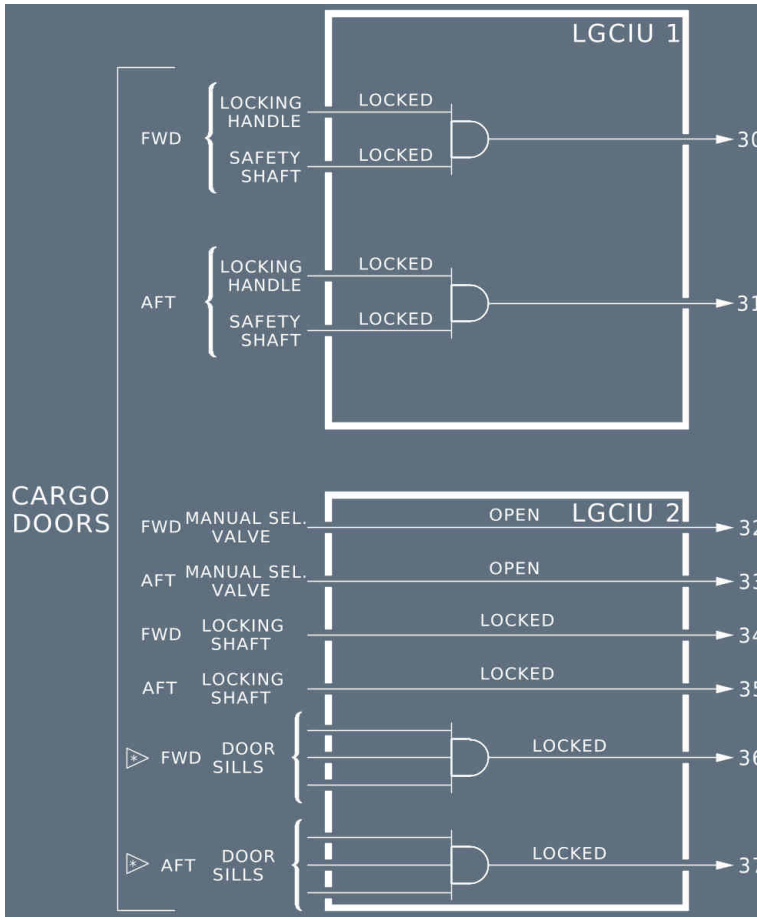
## PROXIMITY DETECTOR OUTPUT SIGNALS

dent.: DSC-32-10-20-00001278.0001001 / 22 MAY 12
**Applicable to: ALL**



## PROXIMITY DETECTOR OUTPUT SIGNALS (CONT'D)

Ident.: DSC-32-10-20-00001279.0001001 / 21 MAR 16
**Applicable to: ALL**



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## GENERAL

Ident.: DSC-32-10-30-00001285.0001001 / 10 DEC 09
**Applicable to: ALL**

The following tables present the operational effects of the proximity detectors on aircraft systems.
How to read the tables :

| SYSTEM | LGCIU 1 | LGCIU 2 | A/C IN FLT | A/C ON GROUND |
|---|---|---|---|---|
| SERVICE INTERPHONE | **6** | **6** | ..... | ..... |
| SFCC 1(2) | **5** | (**5**) | ..... | ..... |

The above lines mean that the service interphone receives the output n° 6 from both LGCIUs, while SFCC 1 receive the output 5 from LGCIU 1 and SFCC 2 the output 5 from LGCIU 2.
The two additional columns give the system functioning when the aircraft is in flight and on the ground.

## PROXIMITY DETECTORS ON SHOCK ABSORBERS

Ident.: DSC-32-10-30-00018608.0001001 / 15 MAY 25
**Applicable to: ALL**

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **GENERAL** | STROBE lights | | **5** | On when AUTO selected | Off when AUTO selected |
| | LOGO lights | | **5** | Off when flaps retracted | On |
| | AIRSTAIRS | **3** | **1** | Control inhibited(1) | Control not inhibited(2) |
| | CARGO DOOR | | **5** | Normal control not available | Normal control available |
| | WATER FILLING | | **5** | Preselect water servicing inhibited | Preselect water servicing available |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **AIR COND** | AVNCS COOLING | **5** | **5** | • Skin temperature < 32 °C : The system is in closed configuration(1)<br>• Skin temperature > 35 °C : The system is in intermediate configuration(1) | • Skin temperature < 9 °C : The system is in closed configuration(2)<br>• Skin temperature > 12 °C : The system is in open configuration(2) |
| | GRND COOLING | **1**<br>**3** | **1**<br>**3** | Inhibited(1) | Not inhibited(2) |
| | FWD CARGO VENT | | **5** | Extract fan stopped when ΔP > 1 PSI | Extract fan on |
| | CAB PRESS | **5** | **5** | Climb mode active(4) | - Prepressurization active before TO (3)<br>- Depressurization active after LDG (3) |
| | PACK 1(2) TEMP CONTROL | | **3**<br>**(1)** | Pack air inlet flaps opened. | Pack air inlet flap fully closed at TO and LDG |
| **APU** | APU AUTO SHUTDOWN | **5** | | In case of oil low pressure, automatic shutdown is delayed by 15.5 s | In case of oil low pressure, the automatic shutdown is delayed by:<br>• 15.5 s if the oil temperature < -4 °C<br>• 0.05 s if oil temperature > -4 °C |
| | APU SPEED CONTROL | | | Speed is controlled at 100 % | Speed is controlled at 99 % (100 % for ENG start or when ambient temperature is below -18 °C or above 35 °C) |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **COMMUNICATIONS** | SERVICE INTERPHONE | **6** | **6** | Inhibited(5) | Available(6) |
| | PUBLIC ADDRESS (PA) | **1**<br>**3** | **1**<br>**3** | PA increased level(5) | PA low level(6) |
| | ADIRU and AVIONICS ground warning | **1**<br>**3** | **1**<br>**3** | External horn and light inhibited(5) | External horn and light not inhibited(6) |
| | FLT INTERPHONE | **9** | | Communication with ground mechanic inhibited | Communication with ground mechanic available |
| | COCKPIT CALL LIGHT | **9** | | Inhibited | Not inhibited |
| | ACARS (ACARS MU or ATSU) | **7** | | Available | Available |
| | CVR | **1**<br>**3**<br>**7** | **1**<br>**3** | Runs(5) | Runs : (6)<br>- During the first 5 min following energization<br>- With at least one engine running<br>Stops : (6)<br>5 min after second engine shutdown |
| | CVR | | **5** | • ERASE function inhibited<br>• No low frequency signal in the loudspeakers if test performed | • ERASE function not inhibited<br>• Low frequency signal in the loudspeakers if test performed |
| **ELEC** | DC generation | **5** | | APU start on batteries only, is delayed by 45 s | No APU start delay when on batteries only |
| | GALLEY supply | | **5** | Main galley not supplied when APU GEN only is supplying | Main galley supplied when APU GEN only is supplying |
| **EIS** | EIS | **5** | | Display test inhibited when ANN LT TEST is selected | Display test not inhibited |
| **FIRE** | APU | **5** | | No APU fire automatic extinguishing | Automatic extinguishing not inhibited |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **FLT CTL** | SFCC 1(2) | **5** | **(5)** | • For SFCC 1(2): Slats alpha/speed lock function active<br>• For SFCC(2): No flaps movement inhibition if the cargo door is opened | • For SFCC 1(2): Slats alpha/speed lock function active if speed > 60 kt<br>• For SFCC (2): Flaps movement inhibition if cargo door is opened |
| **FLT INST** | • DFDR<br>• QAR | **1**<br>**3**<br>**7** | **1**<br>**3** | Runs(5) | Runs: (6)<br>• During the first 5 min following energization<br>• With one engine running<br>Stops: (6)<br>5 min after second engine shut down |
| **FUEL** | FQI | **5** | | FQI uses flight attitude correction due to wing bending | FQI uses ground attitude correction |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **HYD** | BLUE and GREEN pumps | **1**<br>**3** | | Blue or green pump "FAULT" light not inhibited when related pump is stopped(5) | Blue or green pump "FAULT" light inhibited when related pump is stopped(6) |
| | BLUE pump | **7** | | Runs when electrical power is available | Runs when at least one engine is running |
| | BLUE and YELLOW pumps | | **1**<br>**3** | Blue or yellow pump "FAULT" light not inhibited when related pump is stopped(5) | Blue or yellow pump "FAULT" light inhibited when related pump is stopped(7) |
| | PTU | | **7** | PTU runs if green/yellow different pressure > 500 PSI | PTU runs if green/yellow different pressure > 500 PSI and<br>• Both MASTER LEVERS are at OFF or<br>• Both MASTER LEVERS are at ON or<br>• Nose wheel steering is not in towing position with parking brake released.<br>PTU is inhibited during the use of the cargo door hand pump and for 40 s after its use. |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **ICE RAIN PROT** | CAPT, (F/O), ((STBY)) probes and CAPT, (F/O) windshields and windows heating | **4**<br>**(2)**<br>**((8))** | **4**<br>**(2)**<br>**((8))** | • CAPT, (F/O), ((STBY)) pitots and CAPT, (F/O) windshields are heated at high level<br>• All other probes and windows are heated(5) | • With engines stopped: no heating(7)<br>• With at least one engine running:<br>• CAPT, (F/O), ((STBY)) pitots and CAPT, (F/O) windshields are heated at low level(7)<br>• All other probes and windows are heated (except TAT probes). |
| | WING ANTI ICE | **3** | **1** | Wing anti ice valves open when the WING ANTI ICE pb is at ON(5) | Wing anti ice valves open for 30 s when the WING ANTI ICE pb is at ON(7) |
| | RAIN REPELLENT | **1**<br>**3** | **1**<br>**3** | Not inhibited(5) | Inhibited if engines are stopped(6) |
| | DRAIN MAST | | **9** | High heating level is applied | Low heating level is applied |
| **LANDING GEAR** | L/G SAFETY VALVE | **6** | | Safety valve closes if aircraft speed > 264 kt | Safety valve opened |
| | L/G control | **10** | **10** | Retraction not inhibited(8) | Retraction inhibited(8) |
| | TIRE PRESS | | **5** | WHEEL TYRE LO PR threshold set to its flight level | WHEEL TYRE LO PR threshold set to its ground level |
| **NAVIGATION** | STANDBY ALTIMETER | **5** | | VIBRATION function active | VIBRATION function inhibited |
| | ATC 1(2) | **3** | **(1)** | ATC 1(2) available in AUTO mode | ATC 1(2) inhibited in AUTO mode |
| | ADIRU 1 | **7** | | No external horn when ADIRU supplied from batteries only | External horn not inhibited |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | A/C IN FLT | A/C ON GRND |
|---|---|---|---|---|---|
| **POWER PLANT** | FADEC 1(2) | **1**<br>**3**<br>**8** | **(1)**<br>**(3)**<br>**(8)** | On ENG 1(2): (5)<br>• Reverse inhibited<br>• No automatic start abort<br>• FADEC always supplied<br>• FLX not available<br>• If installed, BUMP not selectable. | On ENG 1(2): (7)<br>• Reverse available<br>• Automatic start abort available<br>• 5 min after eng-shut down FADEC 1(2) no more supplied<br>• FLX available<br>• If installed, BUMP selectable. |
| | | **1**<br>**3**<br>**8** | **(1)**<br>**(3)**<br>**(8)** | Modulated idle and approach idle are available(5) | Modulated idle only available(6) |

(1) *When either LGCIU indicates flight.*

(2) *When both LGCIU indicate ground.*

(3) *When either LGCIU indicates ground.*

(4) *When both LGCIU indicate flight.*

(5) *When either output indicates flight.*

(6) *When all outputs indicate ground.*

(7) *When both outputs indicate ground.*

(8) *One valid output is sufficient.*

## PROXIMITY DETECTORS ON UPLOCKS

dent.: DSC-32-10-30-00018610.0002001 / 21 MAR 16
**Applicable to: ALL**

| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | L/G UPLOCKED | L/G NOT UPLOCKED |
|---|---|---|---|---|---|
| **LANDING GEAR** | L/G control | **19** | **19** | If UP selected :(1)<br>L/G doors will close | If UP selected :(1)<br>L/G doors will not close |
| | ECAM WHEEL page | **16**<br>**17**<br>**18** | **16**<br>**17**<br>**18** | If UP selected :(2)<br>L/G uplocked indications | If UP selected :(2)<br>L/G in transit indications |
| | L/G indicator panel | **16**<br>**17**<br>**18** | | If UP selected :(2)<br>no indication | If UP selected :(2)<br>"UNLK" red indications |

(1) *One valid output is sufficient.*

(2) *When all outputs indicate the same position.*

## PROXIMITY DETECTORS ON DOORS

dent.: DSC-32-10-30-00018611.0001001 / 21 MAR 16
**Applicable to: ALL**

| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | DOORS FULLY OPENED | DOORS CLOSED |
|---|---|---|---|---|---|
| **LANDING GEAR** | L/G control | **24** | **24** | L/G extension or retraction possible (1) | L/G extension or retraction inhibited (1) |
| | ECAM WHEEL page | **20**<br>**21**<br>**22**<br>**23** | **20**<br>**21**<br>**22**<br>**23** | Doors fully opened indication | Doors closed indication |

(1) *One valid output is sufficient.*

## PROXIMITY DETECTORS ON DOWNLOCKS

dent.: DSC-32-10-30-00018613.0001001 / 21 MAR 16
**Applicable to: ALL**

| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | L/G DOWNLOCKED | L/G NOT DOWNLOCKED |
|---|---|---|---|---|---|
| **GEN** | TAXI/T.O lights | | **15** | Lights not inhibited | Lights inhibited |
| **COMM** | SIGNS | **12** | **12** | "NO SMOKING" (1)and "EXIT" signs on when AUTO selected(2) | "NO SMOKING" (1)and EXIT" signs inhibited when AUTO selected(3) |

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| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | L/G DOWNLOCKED | L/G NOT DOWNLOCKED |
|---|---|---|---|---|---|
| **FLT INST** | WBS | **15** | **15** | Active (4) | Inhibited (5) |
| **FMGS** | FAC 1(2) | **12** | **12** | VLE indication displayed on PFD 1(2) | No VLE indication |
| **LANDING GEAR** | L/G control | **14** | **14** | If DOWN selected : (6)<br>L/G doors will close | If DOWN selected : (6)<br>L/G doors will not close |
| | ECAM WHEEL page | **11**<br>**13**<br>**15** | **11**<br>**13**<br>**15** | If DOWN selected : (7)<br>L/G down indications | If DOWN selected : (7)<br>L/G in transit indications |
| | L/G INDIC panel | **11**<br>**13**<br>**15** | | If DOWN selected :(7)<br>L/G down indications | If DOWN selected : (7)<br>L/G in transit indications |
| | BRAKING STEERING | **15** | **15** | BSCU test operative(2) | BSCU test inhibited(3) |
| | BRAKE COOLING FANS | | **13** | Cooling available when ON selected | Cooling inhibited when ON selected |
| **NAV** | GPWS | **13** | | "TOO LOW-FLAPS" or "TOO LOW TERRAIN" warning operative | "TOO LOW-GEAR" or "TOO LOW TERRAIN" warning operative |

(1) *For cabin not configured for non smoking flight*

(2) *When either output indicates DOWNLOCK.*

(3) *When both outputs indicate NOT DOWNLOCK.*

(4) *When both outputs indicate DOWNLOCK.*

(5) *When either output indicates NOT DOWNLOCK.*

(6) *One valid output is sufficient.*

(7) *When all outputs indicate the same position.*

## PROXIMITY DETECTORS ON CARGO DOORS

dent.: DSC-32-10-30-00018614.0001001 / 10 MAY 22
**Applicable to: ALL**

### LOCKING HANDLE OR SHAFT, DOOR SILLS

| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | LOCKED | UNLOCKED |
|---|---|---|---|---|---|
| **CRG DOORS** | ECAM DOOR PAGE | **30 (31)** | | Forward (aft) door symbol appears green | Forward (aft) door symbol appears amber, associated with "CARGO" amber |
| | CARGO DOOR OPERATION | | **34 (35)** | Forward (aft) door normal opening inhibition | Forward (aft) door normal opening possible |
| | | | **36 (37)** | Forward (aft) door normal operation possible | Forward (aft) door normal operation inhibited |

### MANUAL SELECTOR VALVE

| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | CLOSE | OPEN |
|---|---|---|---|---|---|
| **CRG DOORS** | CARGO DOOR OPERATION | | **32 (33)** | Forward (aft) door normal opening inhibition | Forward (aft) door normal opening possible |

## PROXIMITY DETECTORS ON FLAPS ATTACHMENTS

dent.: DSC-32-10-30-00001291.0001001 / 22 MAR 16
**Applicable to: ALL**

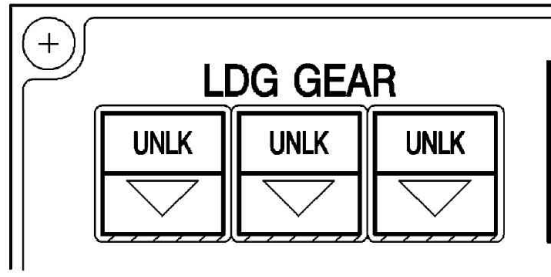
| | SYSTEM | LGCIU 1 OUTPUT | LGCIU 2 OUTPUT | FLAP ATTACHMENT | FLAP ATTACHMENT FAILURE |
|---|---|---|---|---|---|
| **FLT CTL** | SFCC | **28 (29)** | **28 (29)** | L(R) FLAPS normal operation (1) | "FLAPS LOCKED" warning (2) |

(1) *When at least one SFCC detects normal operation*

(2) *When both SFCCs detect attachment failure*

## LANDING GEAR INDICATOR PANEL

Ident.: DSC-32-10-40-00018615.0001001 / 21 MAR 16
**Applicable to: ALL**




This panel is connected to LGCIU1, which receives signals from proximity detectors.

▽ light: comes on green if the gear is locked down.

UNLK light: comes on red if the gear is not locked in the selected position.

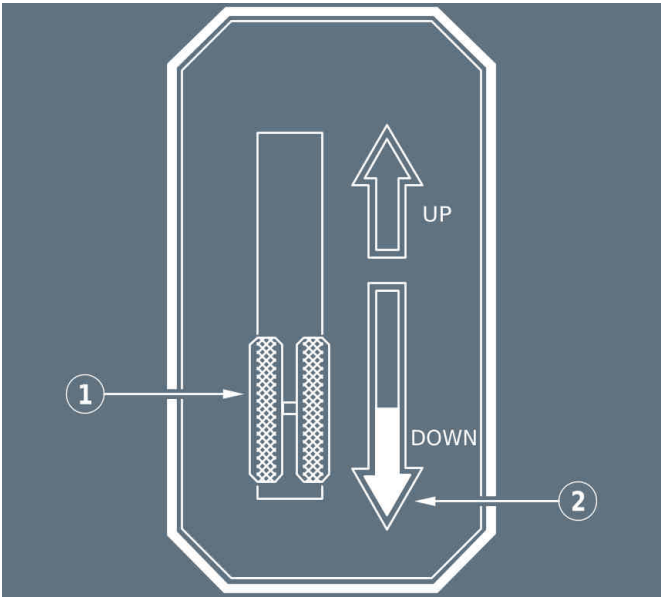
Note: *This panel is connected to the LGCIU1 only, therefore, the lights on the LDG GEAR indicator panel come on as long as the LGCIU1 is electrically supplied.*
*If one UNLK indication remains on, the landing gear position can be confirmed using the WHEEL SD page (information from LGCIU 1 & 2). Only one green triangle on each landing gear is sufficient to confirm that the landing gear is downlocked.*

## LANDING GEAR SELECTOR LEVER

Ident.: DSC-32-10-40-00018616.0001001 / 20 MAY 16
**Applicable to: ALL**

A two-position selector lever sends electrical signals to the two LGCIUs. These control the green hydraulic supply to the landing gear system by means of selector valves.
The flight crew must always move the L/G lever in one continuous movement (i.e. with no stop between both positions).

When the flight crew selects UP or DOWN (and if the airspeed is below 260 kt):
- All landing gear doors open
- Each landing gear moves to the selected position
- All landing gear doors close.




(1) L/G LEVER

UP : This position selects landing gear retraction.
While the landing gear doors are opening, the normal brake system brakes the wheels of the main landing gear automatically.
A brake band in the nose landing gear well brakes the nose landing gear wheels as the doors close (for aircraft equipped with nose landing gear rubbing strips).

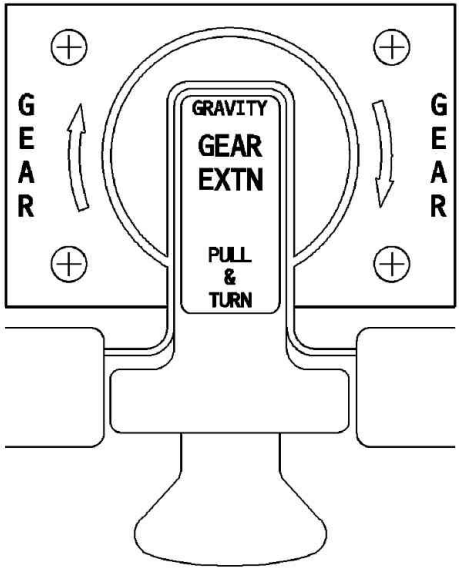
DOWN : This position selects landing gear extension.
An interlock mechanism prevents anyone from accidentally retracting the gear while the aircraft is on the ground. It does so by locking the lever in DOWN position when the shock absorber on either main gear is compressed (aircraft on ground) or the nose wheel steering is not centered.
The landing gear hydraulic system remains pressurized as long as the landing gear is extended (if green hydraulic pressure is available).

(2) RED ARROW

This red arrow lights up if the landing gear is not locked down when the aircraft is in the landing configuration, and a red warning appears on ECAM.

## LANDING GEAR GRAVITY EXTENSION

Ident.: DSC-32-10-40-00018617.0001001 / 21 MAR 16
**Applicable to: ALL**

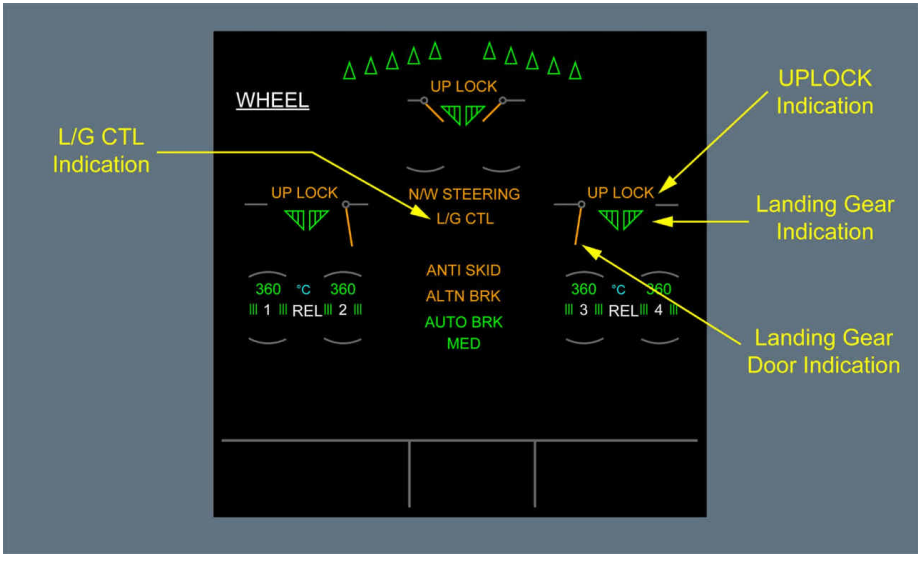



To put the landing gear down by gravity, the flight crew must pull the gear crank out, then turn it clockwise for 3 turns.
When the flight crew operates the crank handle, the cutout valve shuts off hydraulic pressure to the landing gear system and depressurizes it.

## WHEEL SD PAGE

dent.: DSC-32-10-40-00018618.0007001 / 06 NOV 23
**Applicable to: ALL**

Wheel SD Page




### L/G CTL INDICATION

L/G CTL

The associated landing gear position disagrees with the lever position following a 30 s time delay.
The associated ECAM alert is triggered.

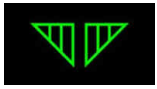
### UPLOCK INDICATION

UP LOCK

The associated landing gear is locked down, and one or more uplocks are in the closed position.
Associated with the following ECAM alert: L/G GEAR UPLOCK FAULT(*Refer to PRO-ABN-LG L/G GEAR UPLOCK FAULT*).

### LANDING GEAR INDICATION

The landing gear is indicated by two triangles for each gear.
The left triangle is controlled by LGCIU 1, the right one by LGCIU 2.



The LGCIU detects the associated landing gear as locked down.

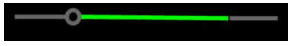
The LGCIU detects the associated landing gear as in transit.

The LGCIU detects the associated landing gear as retracted and uplocked.

The LGCIU is failed.

*Note:* *When the landing gear lever is down, one LGCIU locked down indication (one green triangle per gear) is sufficient to indicate that the landing gear is down and locked.*

## LANDING GEAR DOOR INDICATION



The associated landing gear door is closed and uplocked.

The associated landing gear door is in transit.

The associated landing gear door is open.

## MEMO DISPLAY

Ident.: DSC-32-10-40-00018778.0001001 / 05 MAY 22
**Applicable to: ALL**

FLT L/G DOWN : This memo appears in green if the aircraft is operated in "flight with landing gear down" configuration.

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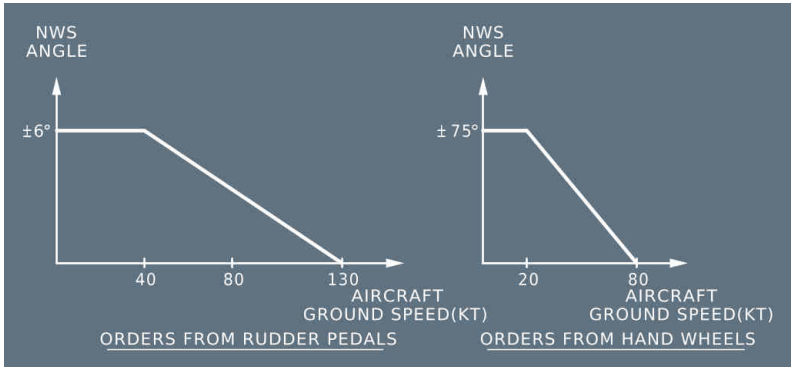
## DESCRIPTION

Ident.: DSC-32-20-10-00018619.0007001 / 28 FEB 18
**Applicable to: ALL**

A hydraulic actuating cylinder steers the nose wheel. The yellow hydraulic system supplies pressure to the cylinder, and electric signals from the Brake and Steering Control Unit (BSCU) control it.

The BSCU receives orders from:
- Captain's, and the First Officer's steering hand wheels (orders added algebraically), or
- Rudder pedals, or
- Autopilot.

The BSCU transforms these orders into nose wheel steering angle. That angle has the following limits, which depend on ground speed and the origin of the orders.



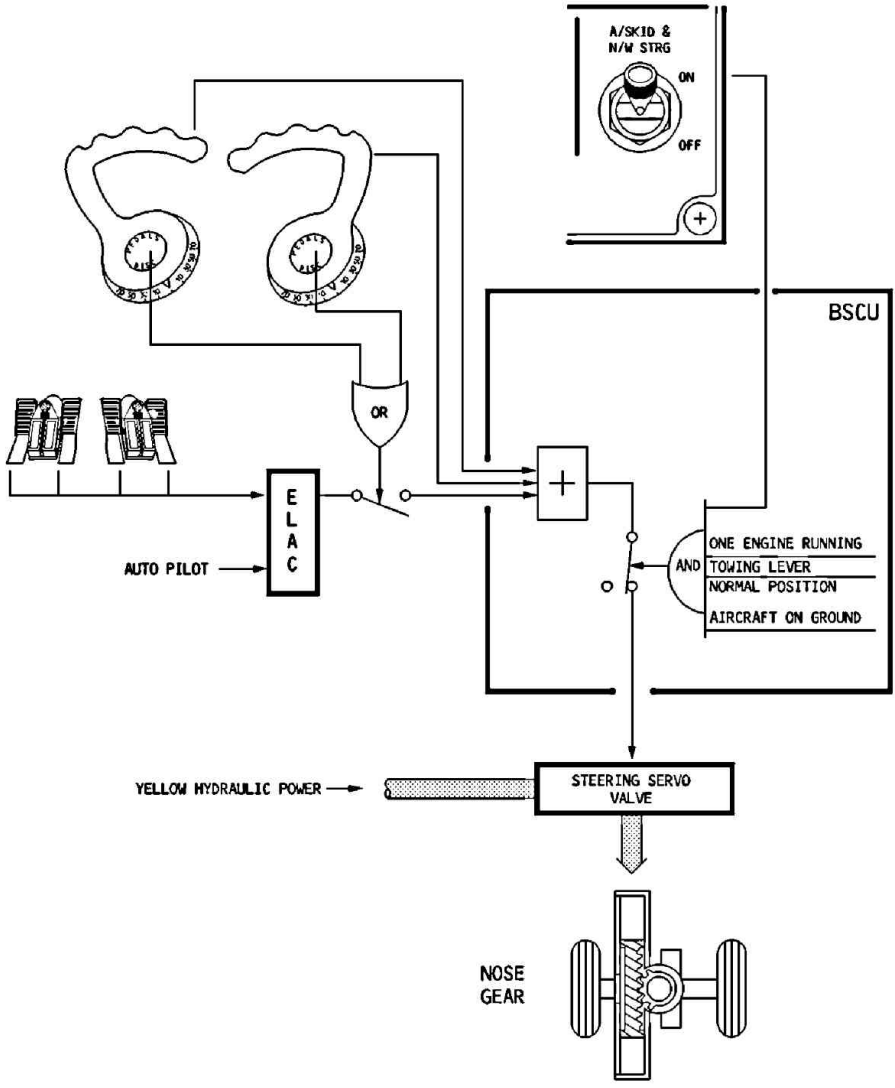

The steering system receives actuating hydraulic pressure when:
- A/SKID & N/W STRG switch is ON
- Towing control lever is in normal position
- At least one engine is running
- Aircraft is on ground.

The handwheel can turn the nosewheel up to 75 ° in either direction. A lever, on the towing electrical box (on nose landing gear), enables ground crew to deactivate the steering system for towing. Then the wheel can be turned 95 ° in either direction.
To prevent rudder pedal orders, or autopilot orders, from going to the BSCU, the pilots can use the pushbutton on either steering handwheel.
An internal cam mechanism returns the nose wheel to the centered position after takeoff.

## ARCHITECTURE

dent.: DSC-32-20-10-00001298.0002001 / 21 MAR 16

**Applicable to: ALL**



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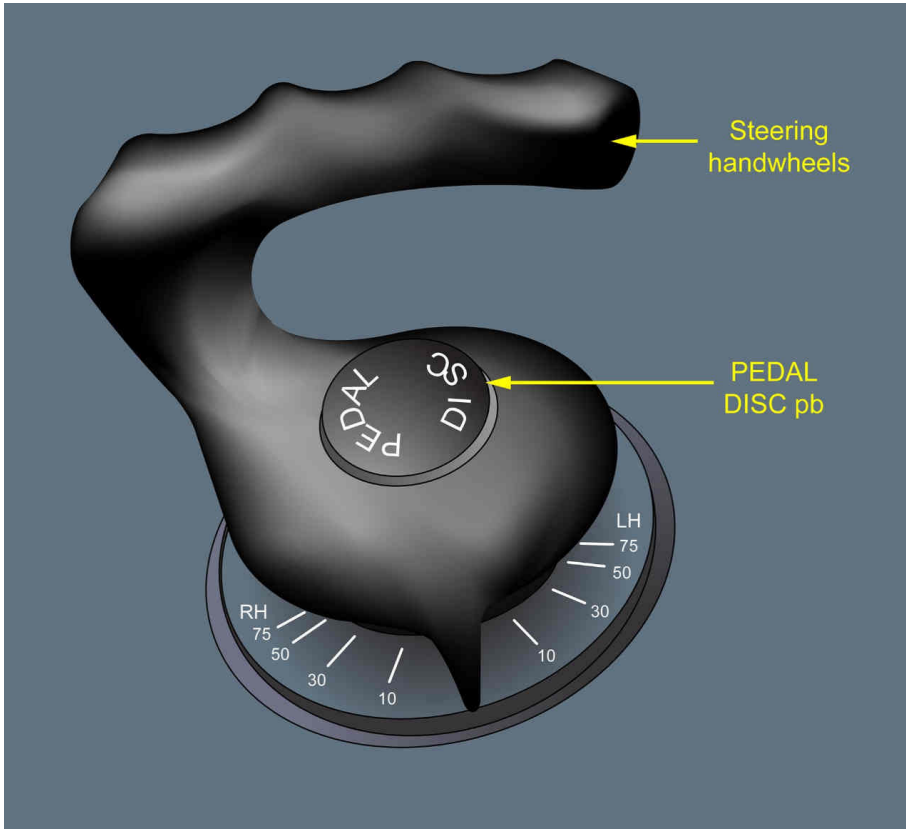
# SIDE CONSOLES

Ident.: DSC-32-20-20-00018620.0001001 / 06 NOV 23
**Applicable to: ALL**

SIDE CONSOLES




## STEERING HANDWHEELS

The steering handwheels, that are interconnected, can steer the nose wheel up to 75 ° in either direction.

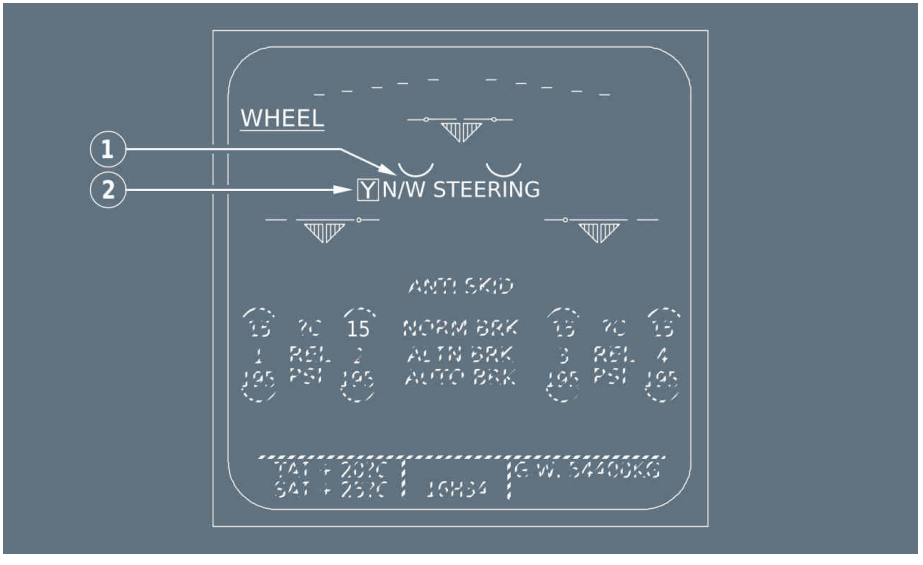
*Note:* *The steering system centers the nose wheel automatically after liftoff.*

## RUDDER PEDAL DISC pb

Pressing this button on either handwheel removes control of nose wheel steering from the rudder pedals until the button is released.

## WHEEL SD PAGE

dent.: DSC-32-20-20-00018622.0004001 / 21 MAR 16
**Applicable to: ALL**




(1) N/W STEERING indication
It appears in amber, when nosewheel steering is lost, due to failure of the nosewheel steering system, or of both BSCU channels, or in case of a yellow hydraulic system low pressure, or if the A/SKID & N/W STRG switch is OFF.

(2) N/W STEERING hydraulic supply indication:

Only when the N/W STEERING indication is displayed, Y is displayed:
- In green when the yellow hydraulic system is not failed, or
- In amber when the yellow hydraulic system low pressure.

## MEMO DISPLAY

Ident.: DSC-32-20-20-00016853.0001001 / 04 JUN 19
**Applicable to: ALL**

NW STRG DISC : This memo appears, when the nose wheel steering selector is in the towing position and with engines not running.

NW STRG DISC : This memo appears, when the nose wheel steering selector is in the towing position and if at least one engine is running.

Intentionally left blank

## GENERAL

Ident.: DSC-32-30-10-00018623.0002001 / 21 MAR 16
**Applicable to: ALL**

The main wheels are equipped with carbon multidisc brakes, which can be actuated by either of two independent brake systems.
The normal system uses green hydraulic pressure, whereas the alternate system uses the yellow hydraulic system backed up by the hydraulic accumulator.
An anti-skid and autobrake system is also provided.
Braking commands come from either the brake pedals (pilot action), or the autobrake system (deceleration rate selected by the crew).
In normal operation, a dual channel Brake and Steering Control Unit (BSCU) controls normal braking and antiskid.

Depending on the failure, braking may revert to:
- Alternate braking with antiskid. This braking mode is controlled by the Alternate Braking Control Unit (ABCU), and the antiskid is controlled by the BSCU
- Alternate braking without antiskid. This braking mode is fully-controlled by the ABCU
- Alternate braking without antiskid on accumulator. This braking mode is fully-controlled by the ABCU.

All the normal and alternate braking components are fully-monitored. Any detected failure is signaled to the crew via ECAM warnings.

The BSCU performs the following secondary functions:
- Checks the residual pressure in the brakes
- Monitors the brake temperature
- Provides discrete wheel speed information to other aircraft systems.

A changeover between the two BSCU channels takes place at each DOWN landing gear lever selection, or in case one channel fails.
The main gear wheels are fitted with fusible plugs which protect against tire burst, in the event of overheat.
Main gear wheels are also equipped with brake cooling fans ◁⊲ , which permit a high speed cooling of brakes.

## ANTI-SKID SYSTEM

Ident.: DSC-32-30-10-00018624.0001001 / 21 MAR 16
**Applicable to: ALL**

The antiskid system provides maximum braking efficiency by maintaining the wheels at the limit of an impending skid.

At skid onset, brake release orders are sent to the normal and alternate servovalves, as well as to the ECAM system which displays the released brakes.
Without using autobrake, full braking performance is achieved only with brake pedals at full deflection.
The antiskid system is deactivated below 20 kt (ground speed).
An ON/OFF switch activates, or deactivates, the antiskid and nosewheel steering systems.

## PRINCIPLE

The speed of each main gear wheel (given by a tachometer) is compared to the aircraft speed (reference speed). When the speed of a wheel decreases below approximately 0.87 times (depending on conditions) reference speed, brake release orders are given to maintain the wheel slip at that value (best braking efficiency).
In normal operation, the reference speed is determined by the BSCU from the horizontal acceleration of ADIRU 1, or ADIRU 2, or ADIRU 3.
In case all ADIRUs fail, reference speed equals the maximum of either main landing gear wheel speeds.

# AUTO BRAKE

dent.: DSC-32-30-10-A-00018625.0001001 / 21 MAR 16
**Applicable to: ALL**

## GENERAL

The purposes of the autobrake system are the following:
- Reduce the braking distance in case of an aborted takeoff
- Establish and maintain a selected deceleration rate during landing, thereby improving passenger comfort and reducing crew workload.

AUTO BRAKE - SYSTEM ARMING

dent.: DSC-32-30-10-A-00018626.0003001 / 09 NOV 17
**Applicable to: MSN 03152, 04656-05136, 07096-12883**

## SYSTEM ARMING

The crew may arm the system by pressing the LO, MED, or MAX pushbutton provided all the following arming conditions are met:
- Green pressure available
- Anti-skid electrically-powered

- No failure in the braking system
- At least one ADIRU is available.

*Note:* *1. Auto brake may be armed with the parking brake on.*
*2. MAX autobrake mode cannot be armed in flight.*

The armed AUTO BRK mode is displayed in blue on the Flight Mode Annunciator (FMA).

Ident.: DSC-32-30-10-A-00018626.0002001 / 21 MAR 16
[2] **Applicable to: MSN 03246-04524, 05276-06737**

## SYSTEM ARMING

The crew may arm the system by pressing the LO, MED, or MAX pushbutton provided all the following arming conditions are met :
- Green pressure available
- Anti-skid electrically-powered
- No failure in the braking system
- At least one ADIRU is available.

*Note:* *1. Auto brake may be armed with the parking brake on.*
*2. MAX autobrake mode cannot be armed in flight.*

AUTO BRAKE - SYSTEM ACTIVATION

Ident.: DSC-32-30-10-A-00018627.0003001 / 09 NOV 17
[3] **Applicable to: MSN 03152, 04656-05136, 07096-12883**

## SYSTEM ACTIVATION

Automatic braking is activated when:
- The command for ground spoilers extension is detected (*Refer to DSC-27-10-20 Speed Brakes and Ground Spoilers - Speed Brake Control*), for LO and MED mode, or
- The command for ground spoilers extension is detected, and the wheel speed is above 40 kt, for MAX mode.

Therefore, if the aircraft makes an acceleration stop and begins to decelerate when the wheel speed is under 72 kt, the automatic braking will not activate because the ground spoilers will not extend.
For autobrake to activate, at least two SEC's must be operative.
The active AUTO BRK mode is displayed in green on the Flight Mode Annunciator (FMA).

dent.: DSC-32-30-10-A-00018627.0002001 / 17 OCT 17
**Applicable to: MSN 03246-04524, 05276-06737**

## SYSTEM ACTIVATION

Automatic braking is activated when:
- The command for ground spoilers extension is detected (*Refer to DSC-27-10-20 Speed Brakes and Ground Spoilers - Speed Brake Control*), for LO and MED mode, or
- The command for ground spoilers extension is detected, and the wheel speed is above 40 kt, for MAX mode.

Therefore, if the aircraft makes an acceleration stop and begins to decelerate when the wheel speed is under 72 kt, the automatic braking will not activate because the ground spoilers will not extend.
For autobrake to activate, at least two SEC's must be operative.

dent.: DSC-32-30-10-A-00018628.0001001 / 21 MAR 16
**Applicable to: ALL**

## SYSTEM DEACTIVATION

The system deactivates when:
- The system disarmed (*Refer to DSC-32-30-10 Auto Brake - System Disarming*), or
- The ground spoilers retract. In this case it remains armed.

dent.: DSC-32-30-10-A-00018629.0002001 / 09 NOV 17
**Applicable to: ALL**

## SYSTEM DISARMING

The system disarms when:
- Flight crew presses the pushbutton switch, or
- One or more arming conditions is lost, or
- After take-off/touch and go, or
- Flight crew applies enough deflection to at least one brake pedal when autobrake is active in MAX, MED or LO mode.

# BRAKING MODES

**Applicable to: ALL**

Ident.: DSC-32-30-10-B-00018630.0001001 / 21 MAR 16

## GENERAL

There are four modes of operation:
- Normal braking
- Alternate braking with antiskid
- Alternate braking without antiskid
- Parking brake.

Ident.: DSC-32-30-10-B-00018631.0001001 / 21 MAR 16

## NORMAL BRAKING

Normal braking is operative when:
- Green hydraulic pressure is available
- A/SKID & N/W STRG switch is ON.

During normal braking, antiskid is operative and autobrake is available.

Braking is electrically-controlled through the BSCU from:
- Pilot's pedals, or
- Automatically activates when:
  - On ground by the autobrake system, or
  - In flight when the landing gear lever is up.

The antiskid system is controlled by the BSCU via the normal servo valves.
There is no brake pressure indication in the cockpit.

Ident.: DSC-32-30-10-B-00018632.0002001 / 02 MAY 24

## ALTERNATE BRAKING WITH ANTI-SKID

Autobrake is inoperative.

Braking uses this mode when green hydraulic pressure is insufficient, and :
- Yellow hydraulic pressure is available
- A/SKID & N/W STRG switch is ON
- Parking brake is not ON.

Braking inputs are made by the brake pedals and sent to the ABCU. Then, taking into account the brake pedal input, the ABCU:
- Energizes the alternate brake selector valve to pressurize the yellow hydraulic circuit
- Electrically controls the Alternate Servo Valve to obtain the correct pressure for the related brakes.

Antiskid is controlled by the BSCU.
A triple indicator on the center instrument panel indicates the pressure delivered to the left and right brakes, as well as the accumulator pressure.

dent.: DSC-32-30-10-B-00018633.0002001 / 02 MAY 24

## ALTERNATE BRAKING WITHOUT ANTI-SKID

Autobrake and antiskid are inoperative.

The antiskid system is either deactivated:
- Electrically (A/SKID & N/W STRG sw OFF, or power supply failure, or BSCU failure), or
- Hydraulically (Y + G system low pressure, the brakes are supplied by the brake accumulator only).

Depending on the brake pedals' demand, the ABCU controls the alternate brake selector and the alternate servovalves.
Brake pressure and accumulator pressure are indicated on a triple indicator, located on the center instrument panel. To avoid wheel locking and limit the risk of tire burst, brake pressure is automatically limited to 1 000 PSI.
The accumulator can supply at least 7 full brake applications.

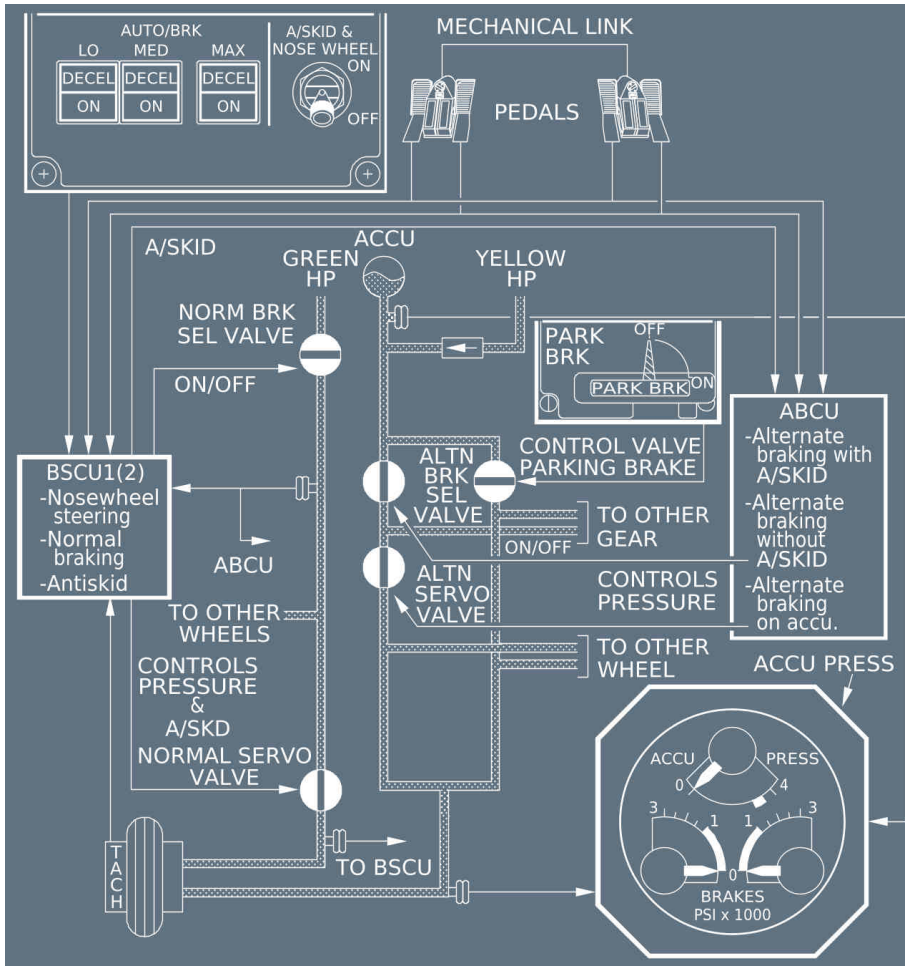
dent.: DSC-32-30-10-B-00018634.0002001 / 21 MAR 16

## PARKING BRAKE

Brakes are supplied by the yellow hydraulic system, or by accumulator pressure via the parking brake control valve, which opens allowing full pressure application on the main gear wheel brakes.
The accumulator maintains the parking pressure for at least 12 h.
If the parking brake is activated and no yellow hydraulic or accumulator brake pressure is available, then the normal braking system can be applied via the brake pedals.
Yellow accumulators can be pressurized by pressing the yellow electrical pump switch.
A triple indicator on the center instrument panel indicates the pressure delivered to the left and right brakes, as well as the accumulator pressure.

## BRAKING SCHEMATIC

Ident.: DSC-32-30-10-00001316.0003001 / 09 OCT 12
**Applicable to: ALL**



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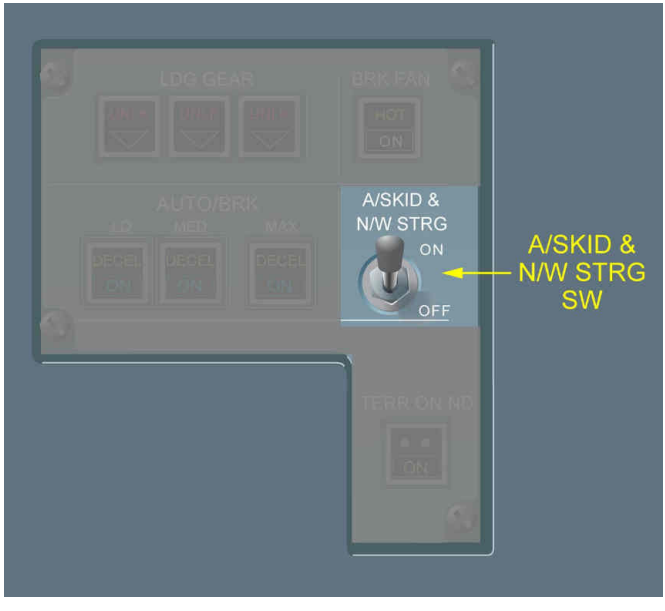
## CENTER INSTRUMENT PANEL

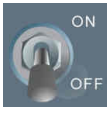
Ident.: DSC-32-30-20-00018635.0002001 / 06 NOV 23
**Applicable to: ALL**

### A/SKID & N/W STRG sw



If green hydraulic pressure is available, then antiskid is available.
If green hydraulic pressure is lost, then:

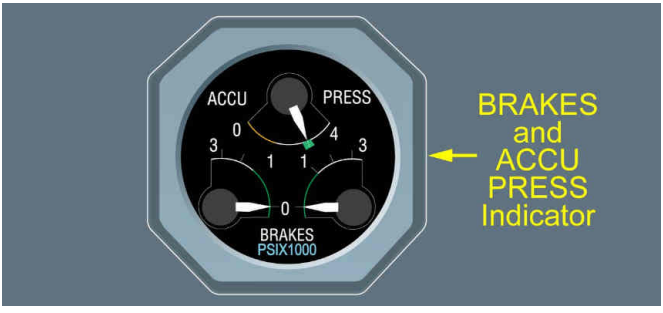
- Yellow hydraulic pressure automatically takes over to supply the brakes
- Antiskid and nosewheel steering remain available
- Triple indicator shows yellow system brake pressure.




The yellow hydraulic system supplies pressure to the brakes.

- Antiskid is deactivated. To avoid wheel locking and limiting the risk of tire burst, brake pressure is automatically limited to 1 000 PSI
- Nosewheel steering is lost
- Differential braking remains available through the pedals
- Triple indicator displays yellow system brake pressure.

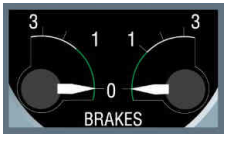
## BRAKES and ACCU PRESS indicator




Brake pressure is only indicated when the yellow hydraulic system controls the brake pressure. This is when the:

- Alternate braking system is activated, or
- Parking brake is applied.

Indicates the pressure in the yellow brake accumulator.

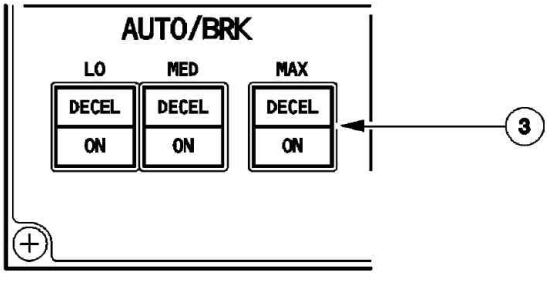



Indicates the yellow pressure delivered to the left and right brakes, as measured downstream of the alternate servovalves.

## AUTO BRK PANEL

dent.: DSC-32-30-20-00018636.0004001 / 21 MAR 16

**Applicable to: MSN 03152-03246, 04468-04806, 05276-05990, 06080-07425, 07614-12883**



(3) AUTO/BRK panel

The springloaded MAX, MED, and LO pushbutton switches arm the appropriate deceleration rate. The usage for each mode are as follow:

- MAX mode is normally selected for takeoff.
  In the case of an aborted takeoff, maximum pressure goes to the brakes, as soon as the system generates the ground spoiler deployment order
- MED or LO mode is normally selected for landing:
  - MED mode sends progressive pressure to the brakes 2 s after the ground spoilers deploy in order to decelerate the aircraft at 3 m/s² (9.8 ft/s²)
  - LO mode sends progressive pressure to the brakes 2 s after the ground spoilers deploy, in order to decelerate the aircraft at 2 m/s² (6.6 ft/s²).

The lighting on the pushbutton switches are as follow:

ON light : comes on blue to indicate positive arming

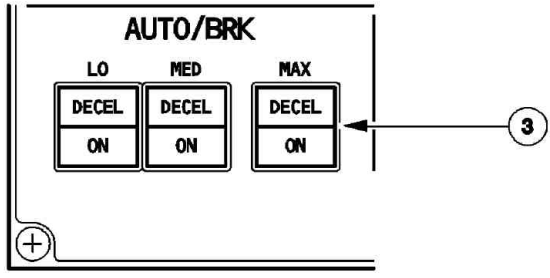
DECEL light : comes on green when the actual deceleration is 80 % of the selected rate

*Note: On slippery runways, the predetermined deceleration may not be reached, due to antiskid operation. In this case, the DECEL light will not come on. This does not mean that autobrake is not working.*

Off : The corresponding autobrake mode is not armed

**AUTO BRK PANEL**

Ident.: DSC-32-30-20-00018636.0002001 / 21 MAR 16
**Applicable to: MSN 04235, 05004-05136, 06072, 07426**



(3) AUTO/BRK panel

The springloaded MAX, MED, and LO pushbutton switches arm the appropriate deceleration rate. The usage for each mode are as follow:

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  In the case of an aborted takeoff, maximum pressure goes to the brakes, as soon as the system generates the ground spoiler deployment order
- MED or LO mode is normally selected for landing:
  - MED mode sends progressive pressure to the brakes 2 s after the ground spoilers deploy in order to decelerate the aircraft at 3 m/s² (9.8 ft/s²)
  - LO mode sends progressive pressure to the brakes 4 s after the ground spoilers deploy, in order to decelerate the aircraft at 1.7 m/s² (5.6 ft/s²).

The lighting on the pushbutton switches are as follow:

ON light : comes on blue to indicate positive arming

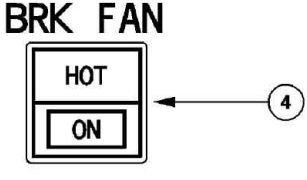
DECEL light : comes on green when the actual deceleration is 80 % of the selected rate

*Note: On slippery runways, the predetermined deceleration may not be reached, due to antiskid operation. In this case, the DECEL light will not come on. This does not mean that autobrake is not working.*

Off : The corresponding autobrake mode is not armed

## BRAKE FAN ⊲

dent.: DSC-32-30-20-00018637.0001001 / 21 MAR 16
**Applicable to: ALL**



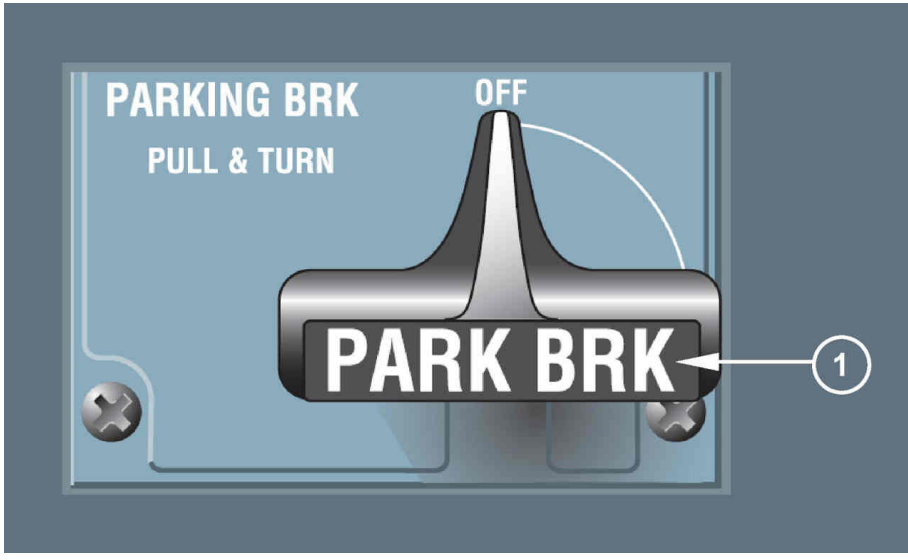

(4) BRK FAN pb-sw ⊲

ON light : The brake fans run if the lefthand main landing gear is down and locked

| | | |
|---|---|---|
| Off | : | The brake fans stop |
| HOT light | : | This amber light comes on when the brakes get too hot (A caution appears on ECAM, also) |

## PEDESTAL

Ident.: DSC-32-30-20-00018638.0001001 / 21 MAR 16
**Applicable to: ALL**

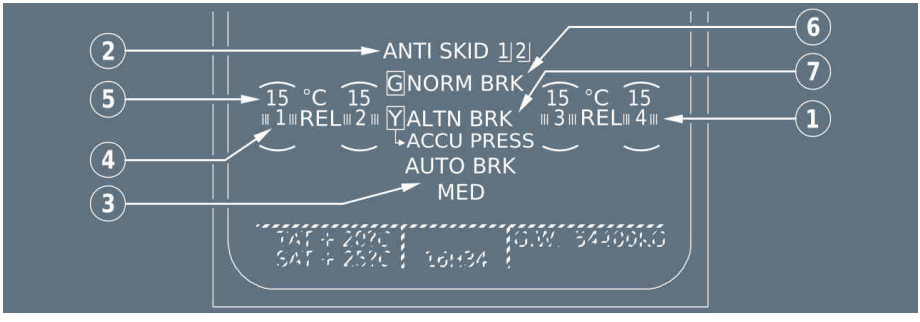



(1) PARKING BRK handle
Flight crew pulls this handle, then turns it clockwise, to apply the parking brake.
The ECAM memo page displays “PARK BRK”.

| | |
|---|---|
| CAUTION | If the pointer is not at ON, the parking brake is not on. |

## WHEEL SD PAGE

dent.: DSC-32-30-20-00018639.0008001 / 21 MAR 16
**Applicable to: ALL**




(1) Release indicators

It appears in amber in case of brake released fault.

(2) ANTI SKID indication

(A) ANTISKID label

It appears in amber, a long with an ECAM caution, in case of a total BSCU failure, or when the A/SKID & N/W STRG sw is OFF, or if the BSCU detects an ANTI-SKID failure, or in case of normal braking and yellow hydraulic system low pressure.
It appears in green in case of autobrake, normal braking, or alternate braking failure, and antiskid is still available.

(B) BSCU channel indication

When ANTISKID label is displayed, the number of the failed system(s) is (are) displayed in amber, if any.

(3) AUTO BRK

This legend appears:
- In green when auto brake is armed, or
- Flashing green for 10 s after autobrake disengagement, or
- In amber, along with an ECAM caution, to indicate a system failure.

MED, LO, or MAX appears underneath in green to show which rate has been selected.

(4) Wheel number
This white number identifies individual wheels of the main landing gear.

(5) Brake temperature

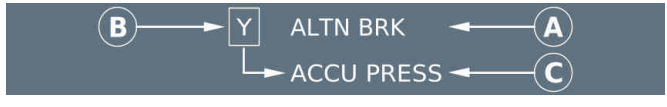
- Temperature normally appears in green.
- Green arc appears on the hottest wheel when one brake temperature exceeds 100 °C.
- Green arc becomes amber, and an ECAM caution appears, when the corresponding brake temperature exceeds 300 °C.

(6) NORM BRK indication

(A) NORM BRK label
This indication appears in green when autobrake or alternate braking is failed, and normal braking is still available.
The legend appears in amber when normal braking is failed due to total BSCU failure, or to the loss of the green hydraulic pressure, or to the loss of antiskid.

(B) NORM BRK hydraulic supply indication
G is displayed when the NORM BRK label is displayed. It is green when green hydraulic pressure is available and amber, in case of green hydraulic low pressure.

(7)




(A) ALTN BRK label
This indication appears in green, if the braking system is in alternate mode and not failed, or in case autobrake or normal braking is failed and alternate braking is still available.
This indication appears in amber when alternate braking is failed.

(B) ALTN BRK hydraulic supply indication
Y is displayed when the ALTN BRK label is displayed. It is green when yellow hydraulic pressure is available and amber, in case of yellow hydraulic low pressure.

(C) ACCU indications

(*) Appears in green, when the ALTN BRK label is displayed, and the yellow hydraulic pressure is available.
Appears in amber, with no arrow, when the yellow hydraulic system and the accumulator are in low pressure.

(**) Appears in green when the alternate braking is pressurized by the yellow accumulator.

## AUDIO INDICATORS

dent.: DSC-32-30-20-00025022.0001001 / 02 MAY 24
**Applicable to: MSN 03152-05004, 05063-05889, 06080-07425, 07614-12883**

| Audio Indicator | Condition or Cause | Duration |
|---|---|---|
| Auto Brake Off (Synthetic Voice) | The flight crew disconnects the autobrake at a speed above 33 kt, after touchdown or after RTO.<br>Associated with the AUTO BRK OFF memo.<br>The master caution light comes on for 3 s and a single chime sounds. | Sounds one time |
| Auto Brake Off, Auto Brake Off (Synthetic Voice) | The autobrake is disconnected due to a system failure after touchdown or after RTO.<br>Associated with the AUTO BRK OFF memo.<br>The master caution light comes on and a single chime sounds. | Sounds one time |

## MEMO DISPLAY

dent.: DSC-32-30-20-A-00016854.0001001 / 22 MAR 16
**Applicable to: ALL**

AUTO BRK LO/MED/MAX : This memo appears in green, depending on the selection of the AUTO BRK pb.

AUTO BRK OFF

Ident.: DSC-32-30-20-A-00016855.0002001 / 09 NOV 21
3 **Applicable to: MSN 03152-05004, 05063-05889, 06080-07425, 07614-12883**

AUTO BRK OFF — Appears if:
- The flight crew disconnects the autobrake, or
- The autobrake fails.

Ident.: DSC-32-30-20-A-00016855.0001001 / 22 MAR 16
**Applicable to: MSN 05058, 05990-06072, 07426**

AUTO BRK OFF : This memo appears in green if the auto brake is failed.

Ident.: DSC-32-30-20-A-00016856.0001001 / 22 MAR 16
**Applicable to: ALL**

BRK FAN : This memo appears in green if the BRK FAN pb is ON.

Ident.: DSC-32-30-20-A-00016857.0001001 / 22 MAR 16
**Applicable to: ALL**

PARK BRK : This memo appears in green, if the parking brake is ON, during flight phases 1, 2, 9 and 10.

Intentionally left blank